

DBS - JOINS AND SET OPERATIONS

```
mysql> USE sec13;
Database changed
mysql> DROP TABLE IF EXISTS class;
Query OK, 0 rows affected (0.06 sec)

mysql> DROP TABLE IF EXISTS class_info;
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE class (
  ->     id INT,
  ->     name VARCHAR(30)
  -> );
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE class_info (
  ->     id INT,
  ->     address VARCHAR(30)
  -> );
Query OK, 0 rows affected (0.05 sec)

mysql> INSERT INTO class VALUES
  -> (1,'Ananya'),
  -> (2,'Akhil'),
  -> (4,'Bala');
Query OK, 3 rows affected (0.03 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO class_info VALUES
  -> (1,'DELHI'),
  -> (2,'MUMBAI'),
  -> (3,'CHENNAI');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT *
-> FROM class
-> CROSS JOIN class_info;
+-----+-----+-----+-----+
| id | name | id | address |
+-----+-----+-----+-----+
| 1 | Ananya | 1 | DELHI |
| 2 | Akhil | 1 | DELHI |
| 4 | Bala | 1 | DELHI |
| 1 | Ananya | 2 | MUMBAI |
| 2 | Akhil | 2 | MUMBAI |
| 4 | Bala | 2 | MUMBAI |
| 1 | Ananya | 3 | CHENNAI |
| 2 | Akhil | 3 | CHENNAI |
| 4 | Bala | 3 | CHENNAI |
+-----+-----+-----+-----+
9 rows in set (0.01 sec)
```

```
mysql> INSERT INTO class VALUES
-> (1, 'Ananya'),
-> (2, 'Akhil'),
-> (4, 'Bala');
Query OK, 3 rows affected (0.02 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO class_info VALUES
-> (1, 'DELHI'),
-> (2, 'MUMBAI'),
-> (3, 'CHENNAI');
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM class;
```

id	name
1	Ananya
2	Akhil
4	Bala
1	Ananya
2	Akhil
4	Bala

```
6 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM class_info;
```

id	address
1	DELHI
2	MUMBAI
3	CHENNAI
1	DELHI
2	MUMBAI
3	CHENNAI

```
6 rows in set (0.00 sec)
```

```
mysql> SELECT *
-> FROM class
-> CROSS JOIN class_info;
```

id	name	id	address
1	Ananya	1	DELHI
2	Akhil	1	DELHI
4	Bala	1	DELHI
1	Ananya	1	DELHI
2	Akhil	1	DELHI
4	Bala	1	DELHI
1	Ananya	2	MUMBAI
2	Akhil	2	MUMBAI
4	Bala	2	MUMBAI
1	Ananya	2	MUMBAI
2	Akhil	2	MUMBAI
4	Bala	2	MUMBAI
1	Ananya	3	CHENNAI
2	Akhil	3	CHENNAI
4	Bala	3	CHENNAI
1	Ananya	3	CHENNAI
2	Akhil	3	CHENNAI
4	Bala	3	CHENNAI
1	Ananya	1	DELHI
2	Akhil	1	DELHI
4	Bala	1	DELHI
1	Ananya	1	DELHI
2	Akhil	1	DELHI
4	Bala	1	DELHI
1	Ananya	2	MUMBAI
2	Akhil	2	MUMBAI
4	Bala	2	MUMBAI
1	Ananya	2	MUMBAI
2	Akhil	2	MUMBAI
4	Bala	2	MUMBAI
1	Ananya	3	CHENNAI
2	Akhil	3	CHENNAI
4	Bala	3	CHENNAI
1	Ananya	3	CHENNAI
2	Akhil	3	CHENNAI

36 rows in set (0.00 sec)

QUESTION 1 - CROSS JOIN

```
mysql> USE sec13;
Database changed
mysql> SELECT * FROM class;
+-----+-----+
| id  | name  |
+-----+-----+
| 1   | Ananya |
| 2   | Akhil  |
| 4   | Bala   |
| 1   | Ananya |
| 2   | Akhil  |
| 4   | Bala   |
+-----+-----+
6 rows in set (0.00 sec)

mysql> SELECT * FROM class_info;
+-----+-----+
| id  | address |
+-----+-----+
| 1   | DELHI   |
| 2   | MUMBAI  |
| 3   | CHENNAI |
| 1   | DELHI   |
| 2   | MUMBAI  |
| 3   | CHENNAI |
+-----+-----+
6 rows in set (0.00 sec)
```

```
mysql> SELECT *
-> FROM class
-> CROSS JOIN class_info;
```

id	name	id	address
1	Ananya	1	DELHI
2	Akhil	1	DELHI
4	Bala	1	DELHI
1	Ananya	1	DELHI
2	Akhil	1	DELHI
4	Bala	1	DELHI
1	Ananya	2	MUMBAI
2	Akhil	2	MUMBAI
4	Bala	2	MUMBAI
1	Ananya	2	MUMBAI
2	Akhil	2	MUMBAI
4	Bala	2	MUMBAI
1	Ananya	3	CHENNAI
2	Akhil	3	CHENNAI
4	Bala	3	CHENNAI
1	Ananya	3	CHENNAI
2	Akhil	3	CHENNAI
4	Bala	3	CHENNAI
1	Ananya	1	DELHI
2	Akhil	1	DELHI

1	Ananya	1	DELHI
2	Akhil	1	DELHI
4	Bala	1	DELHI
1	Ananya	1	DELHI
2	Akhil	1	DELHI
4	Bala	1	DELHI
1	Ananya	2	MUMBAI
2	Akhil	2	MUMBAI
4	Bala	2	MUMBAI
1	Ananya	2	MUMBAI
2	Akhil	2	MUMBAI
4	Bala	2	MUMBAI
1	Ananya	3	CHENNAI
2	Akhil	3	CHENNAI
4	Bala	3	CHENNAI
1	Ananya	3	CHENNAI
2	Akhil	3	CHENNAI
4	Bala	3	CHENNAI

```
36 rows in set (0.00 sec)
```

PART 2: INNER JOIN

```
mysql> SELECT *  
      -> FROM class  
      -> INNER JOIN class_info  
      -> ON class.id = class_info.id;
```

id	name	id	address
1	Ananya	1	DELHI
1	Ananya	1	DELHI
2	Akhil	2	MUMBAI
2	Akhil	2	MUMBAI
1	Ananya	1	DELHI
1	Ananya	1	DELHI
2	Akhil	2	MUMBAI
2	Akhil	2	MUMBAI

```
8 rows in set (0.01 sec)
```

QUESTION 2 - INNER JOIN

```
mysql> SELECT *
-> FROM class
-> INNER JOIN class_info
-> ON class.id = class_info.id;
```

id	name	id	address
1	Ananya	1	DELHI
1	Ananya	1	DELHI
2	Akhil	2	MUMBAI
2	Akhil	2	MUMBAI
1	Ananya	1	DELHI
1	Ananya	1	DELHI
2	Akhil	2	MUMBAI
2	Akhil	2	MUMBAI

8 rows in set (0.01 sec)

QUESTION 3 - INNER JOIN

```
mysql> SELECT class.name,
-> class_info.address
-> FROM class
-> INNER JOIN class_info
-> ON class.id = class_info.id;
```

name	address
Ananya	DELHI
Ananya	DELHI
Akhil	MUMBAI
Akhil	MUMBAI
Ananya	DELHI
Ananya	DELHI
Akhil	MUMBAI
Akhil	MUMBAI

8 rows in set (0.00 sec)

PART 3: NATURAL JOIN

```
mysql> SELECT *  
      -> FROM class  
      -> NATURAL JOIN class_info;  
+-----+-----+-----+  
| id  | name  | address |  
+-----+-----+-----+  
| 1   | Ananya | DELHI   |  
| 1   | Ananya | DELHI   |  
| 2   | Akhil  | MUMBAI  |  
| 2   | Akhil  | MUMBAI  |  
| 1   | Ananya | DELHI   |  
| 1   | Ananya | DELHI   |  
| 2   | Akhil  | MUMBAI  |  
| 2   | Akhil  | MUMBAI  |  
+-----+-----+-----+  
8 rows in set (0.02 sec)
```

PART 4: LEFT OUTER JOIN

```
mysql> INSERT INTO class VALUES  
      -> (5, 'ashish');  
Query OK, 1 row affected (0.04 sec)  
  
mysql> INSERT INTO class_info VALUES  
      -> (7, 'NOIDA'),  
      -> (8, 'PANIPAT');  
Query OK, 2 rows affected (0.01 sec)  
Records: 2  Duplicates: 0  Warnings: 0
```

QUESTION 5 - LEFT JOIN

```
mysql> SELECT *
-> FROM class
-> LEFT OUTER JOIN class_info
-> ON class.id = class_info.id;
```

id	name	id	address
1	Ananya	1	DELHI
1	Ananya	1	DELHI
2	Akhil	2	MUMBAI
2	Akhil	2	MUMBAI
4	Bala	NULL	NULL
1	Ananya	1	DELHI
1	Ananya	1	DELHI
2	Akhil	2	MUMBAI
2	Akhil	2	MUMBAI
4	Bala	NULL	NULL
5	ashish	NULL	NULL

11 rows in set (0.01 sec)

QUESTION 6 - LEFT JOIN

```
mysql> SELECT *
-> FROM class
-> LEFT JOIN class_info
-> ON class.id = class_info.id
-> WHERE class_info.id IS NULL;
```

id	name	id	address
4	Bala	NULL	NULL
4	Bala	NULL	NULL
5	ashish	NULL	NULL

3 rows in set (0.01 sec)

PART 5: RIGHT OUTER JOIN

QUESTION 7 - RIGHT JOIN

```
mysql> SELECT *
-> FROM class
-> RIGHT OUTER JOIN class_info
-> ON class.id=class_info.id;
```

id	name	id	address
1	Ananya	1	DELHI
1	Ananya	1	DELHI
2	Akhil	2	MUMBAI
2	Akhil	2	MUMBAI
NULL	NULL	3	CHENNAI
1	Ananya	1	DELHI
1	Ananya	1	DELHI
2	Akhil	2	MUMBAI
2	Akhil	2	MUMBAI
NULL	NULL	3	CHENNAI
NULL	NULL	7	NOIDA
NULL	NULL	8	PANIPAT

12 rows in set (0.00 sec)

QUESTION 8 - RIGHT JOIN

```
mysql> SELECT *
-> FROM class
-> RIGHT JOIN class_info
-> ON class.id=class_info.id
-> WHERE class.id IS NULL;
```

id	name	id	address
NULL	NULL	3	CHENNAI
NULL	NULL	3	CHENNAI
NULL	NULL	7	NOIDA
NULL	NULL	8	PANIPAT

4 rows in set (0.00 sec)

PART 6: FULL OUTER JOIN

QUESTION 9 - FULL OUTER JOIN

```
mysql> SELECT *  
-> FROM class  
-> LEFT JOIN class_info  
-> ON class.id = class_info.id  
->  
-> UNION  
->  
-> SELECT *  
-> FROM class  
-> RIGHT JOIN class_info  
-> ON class.id = class_info.id;
```

id	name	id	address
1	Ananya	1	DELHI
2	Akhil	2	MUMBAI
4	Bala	NULL	NULL
5	ashish	NULL	NULL
NULL	NULL	3	CHENNAI
NULL	NULL	7	NOIDA
NULL	NULL	8	PANIPAT

7 rows in set (0.04 sec)

QUESTION 10 - FULL OUTER JOIN

```
mysql> SELECT *
-> FROM class
-> LEFT JOIN class_info
-> ON class.id = class_info.id
-> WHERE class_info.id IS NULL
->
-> UNION
->
-> SELECT *
-> FROM class
-> RIGHT JOIN class_info
-> ON class.id = class_info.id
-> WHERE class.id IS NULL;
+-----+-----+-----+-----+
| id    | name  | id    | address |
+-----+-----+-----+-----+
| 4     | Bala  | NULL  | NULL    |
| 5     | ashish | NULL  | NULL    |
| NULL  | NULL  | 3     | CHENNAI |
| NULL  | NULL  | 7     | NOIDA   |
| NULL  | NULL  | 8     | PANIPAT |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

PART 7: UNION

```

mysql> CREATE TABLE first_table (
    ->     id INT,
    ->     name VARCHAR(30)
    -> );
Query OK, 0 rows affected (0.16 sec)

mysql> CREATE TABLE second_table (
    ->     id INT,
    ->     name VARCHAR(30)
    -> );
Query OK, 0 rows affected (0.08 sec)

mysql> INSERT INTO first_table VALUES
    -> (1,'abhi'),
    -> (2,'adam');
Query OK, 2 rows affected (0.03 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> INSERT INTO second_table VALUES
    -> (2,'adam'),
    -> (3,'chester');
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM first_table;
+-----+-----+
| id    | name  |
+-----+-----+
| 1     | abhi  |
| 2     | adam  |
+-----+-----+
2 rows in set (0.00 sec)

mysql> SELECT * FROM second_table;
+-----+-----+
| id    | name  |
+-----+-----+
| 2     | adam  |
| 3     | chester |
+-----+-----+

mysql> SELECT * FROM second_table;
+-----+-----+
| id    | name  |
+-----+-----+
| 2     | adam  |
| 3     | chester |
+-----+-----+
2 rows in set (0.00 sec)

```

QUESTION 11 - UNION

```
mysql> SELECT * FROM first_table
-> UNION
-> SELECT * FROM second_table;
+-----+-----+
| id    | name    |
+-----+-----+
| 1     | abhi    |
| 2     | adam    |
| 3     | chester |
+-----+-----+
3 rows in set (0.00 sec)
```

QUESTION 12 - UNION

```
mysql> SELECT name FROM first_table
-> UNION
-> SELECT name FROM second_table;
+-----+
| name    |
+-----+
| abhi    |
| adam    |
| chester |
+-----+
3 rows in set (0.00 sec)
```

PART 8: UNION ALL

QUESTION 13 - UNION ALL

```
mysql> SELECT * FROM first_table
-> UNION ALL
-> SELECT * FROM second_table;
+-----+-----+
| id    | name    |
+-----+-----+
| 1     | abhi    |
| 2     | adam    |
| 2     | adam    |
| 3     | chester |
+-----+-----+
4 rows in set (0.00 sec)
```

QUESTION 14 - UNION ALL

```
mysql> SELECT COUNT(*)
-> FROM
-> (
-> SELECT * FROM first_table
-> UNION ALL
-> SELECT * FROM second_table
-> ) A;
+-----+
| COUNT(*) |
+-----+
|         4 |
+-----+
1 row in set (0.02 sec)
```

PART 9: INTERSECT

QUESTION 15 - INTERSECT

```
mysql> SELECT *
-> FROM first_table
-> WHERE (id, name) IN (
->     SELECT id, name
->     FROM second_table
-> );
+-----+-----+
| id  | name |
+-----+-----+
|    2 | adam |
+-----+-----+
1 row in set (0.01 sec)
```

QUESTION 16 - INTERSECT

```
mysql> SELECT name
-> FROM first_table
-> WHERE name IN (
->     SELECT name
->     FROM second_table
-> );
+-----+
| name |
+-----+
| adam |
+-----+
1 row in set (0.00 sec)
```


PART 10: MINUS

QUESTION 17 - MINUS

```
mysql> SELECT *
-> FROM first_table
-> WHERE NOT EXISTS (
->     SELECT 1
->     FROM second_table
->     WHERE second_table.id = first_table.id
->     AND second_table.name = first_table.name
-> );
+-----+-----+
| id    | name |
+-----+-----+
| 1     | abhi |
+-----+-----+
1 row in set (0.01 sec)
```

QUESTION 18 - MINUS

```
mysql> SELECT name
-> FROM first_table
-> WHERE name NOT IN (
->     SELECT name
->     FROM second_table
-> );
+-----+
| name |
+-----+
| abhi |
+-----+
1 row in set (0.02 sec)
```

QUESTION 19

```
mysql> SELECT c.id, c.name, ci.address  
-> FROM class c  
-> INNER JOIN class_info ci  
-> ON c.id = ci.id;
```

id	name	address
1	Ananya	DELHI
1	Ananya	DELHI
2	Akhil	MUMBAI
2	Akhil	MUMBAI
1	Ananya	DELHI
1	Ananya	DELHI
2	Akhil	MUMBAI
2	Akhil	MUMBAI

8 rows in set (0.02 sec)

QUESTION 20

```
mysql> SELECT c.id,
->          c.name,
->          CASE
->              WHEN ci.address IS NULL
->              THEN 'Address Missing'
->              ELSE 'Address Available'
->          END AS Status
-> FROM class c
-> LEFT JOIN class_info ci
-> ON c.id = ci.id;
```

id	name	Status
1	Ananya	Address Available
1	Ananya	Address Available
2	Akhil	Address Available
2	Akhil	Address Available
4	Bala	Address Missing
1	Ananya	Address Available
1	Ananya	Address Available
2	Akhil	Address Available
2	Akhil	Address Available
4	Bala	Address Missing
5	ashish	Address Missing

11 rows in set (0.01 sec)